

Networked & Distributed Control: Assignment 4

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Abstract—This report addresses the distributed consensus problem over multi-agent systems using centralized and decentralized optimization techniques. Several iterative methods—including Constant Step Size (CSS), Exact Line Search (ELC), Accelerated Nesterov Method (ANM), Combined Consensus (CC), and Alternating Direction Method of Multipliers (ADMM)—are implemented and compared to a centralized baseline. The goal is for all agents to agree on a common terminal state while optimizing individual trajectories. Performance is evaluated in terms of convergence speed and deviation from the centralized solution.

I. INTRODUCTION

This assignment explores distributed optimization in a multi-agent setting, where agents aim to reach a common target state through local dynamics and control input constraints. Each agent has distinct linear dynamics represented by (A_i, B_i) matrices and initial states. The centralized optimal control problem is first formulated and solved. Then, several decentralized algorithms are developed and benchmarked.

The key focus is on subgradient-based optimization, dual decomposition, and consensus averaging, allowing agents to iteratively refine their terminal states and converge to a shared solution.

II. PROBLEM SETUP

A. System Model

The system comprises $N = 4$ agents, each with linear time-invariant dynamics:

$$x_{i,k+1} = A_i x_{i,k} + B_i u_{i,k}, \quad i = 1, 2, 3, 4$$

Initial states $x_{i,0}$ are provided via the dataset `agents.mat`. The prediction horizon N_p and maximum input bound $u_{\max}$ are also given.

B. Objective

The goal is to drive all agents to a common terminal state x_f at time N_p , while minimizing:

$$J = \sum_{i=1}^N \sum_{k=0}^{N_p-1} (\|x_{i,k}\|^2 + \|u_{i,k}\|^2) + \|x_{i,N_p}\|^2$$

subject to:

- Dynamics: $x_{i,k+1} = A_i x_{i,k} + B_i u_{i,k}$
- Input constraints: $|u_{i,k}| \leq u_{\max}$
- Terminal constraint: $x_{i,N_p} = x_f, \quad \forall i$

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III. CENTRALIZED SOLUTION

A centralized convex optimization problem is formulated using CVXPY. The global optimal x_f and control inputs u_i are obtained for all agents. This serves as the performance baseline.

IV. DECENTRALIZED ALGORITHMS

A. Constant Step Size (CSS)

Each agent solves its own optimization problem using a fixed Lagrange multiplier λ_i . Subgradients are computed from deviations from the mean terminal state:

$$\lambda_i^{k+1} = \lambda_i^k + \alpha (x_{i,N_p}^k - \bar{x}_f^k)$$

where $\bar{x}_f^k = \frac{1}{N} \sum_i x_{i,N_p}^k$.

B. Exact Line Search (ELC)

Instead of using a fixed step size, α is chosen optimally at each iteration:

$$\alpha^k = \frac{x^\top g}{g^\top g}, \quad \text{where } g = \text{subgradient}$$

C. Accelerated Nesterov Method (ANM)

ANM introduces a momentum term η to improve convergence speed [1]:

$$\lambda^{k+1} = \eta^k + \alpha^k g^k, \quad \eta^{k+1} = \lambda^{k+1} + \mu(\lambda^{k+1} - \lambda^k)$$

D. Combined Consensus (CC)

A consensus averaging step is introduced:

$$x_i^{r+1} = x_i^r + \sum_j W_{ij} (x_j^r - x_i^r)$$

after which subgradients are calculated and Lagrange multipliers are updated.

E. ADMM

ADMM decomposes the problem with augmented Lagrangian terms[2]:

$$J = J_i + \lambda_i^\top (x_i - z) + \frac{\rho}{2} \|x_i - z\|^2$$

The shared state z is updated via:

$$z = \frac{1}{N} \sum_i (x_i + \frac{1}{\rho} \lambda_i)$$

V. RESULTS

A. Convergence Comparison

Figure 1 compares convergence of all methods using the norm $\|x_i - x_f\|$:

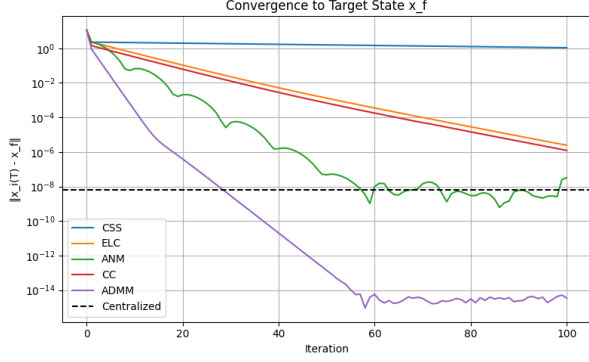


Fig. 1. Convergence to common terminal state x_f (log scale).

B. State Trajectories

For each method, the evolution of x_i across iterations is visualized:

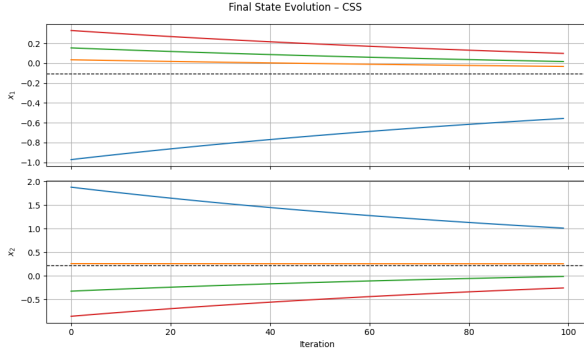


Fig. 2. State evolution of agents using dual decomposition of subgradient descent using constant alpha.

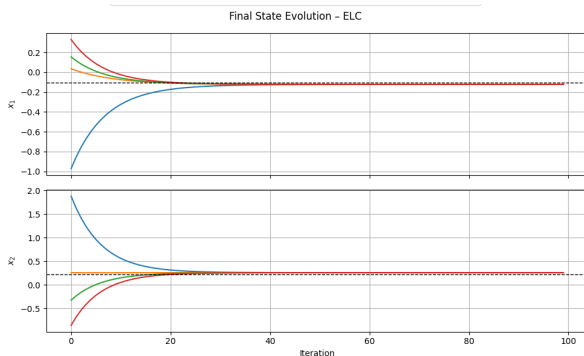


Fig. 3. State evolution of agents using Exact Line Search.

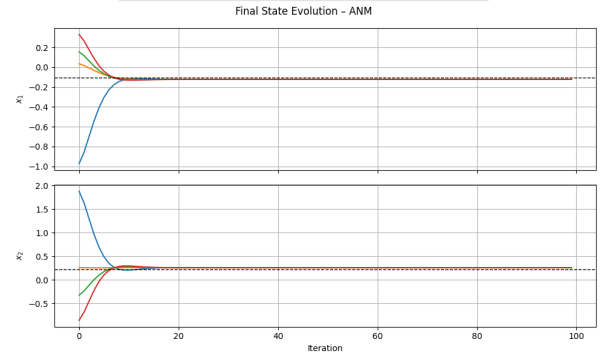


Fig. 4. State evolution of agents using Accelerated Nesterov Method.

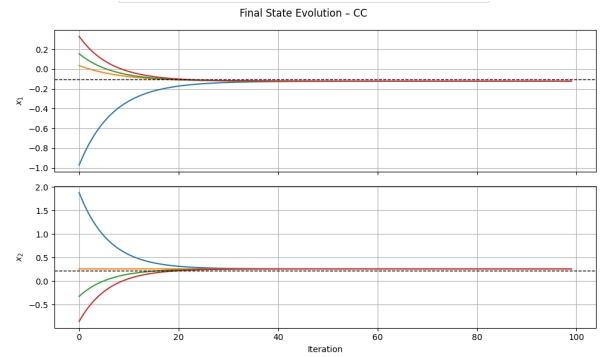


Fig. 5. State evolution of agents using combined consensus.

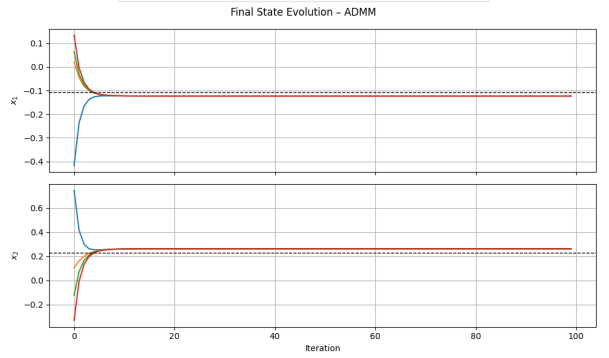


Fig. 6. State evolution of agents using ADMM.

C. Parameter Tuning

The performance of CC and ADMM is evaluated under varying internal parameters (number of consensus steps or ρ). Optimal values are selected based on subgradient convergence.

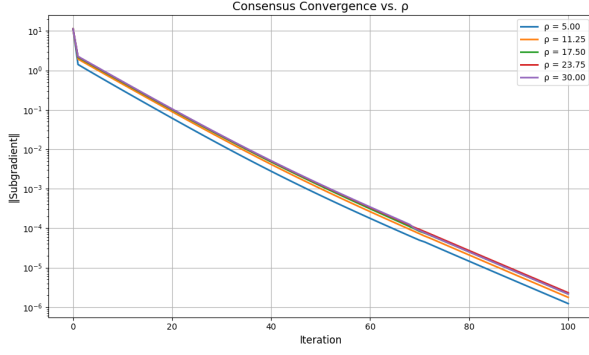


Fig. 7. State evolution of agents using combined consensus.

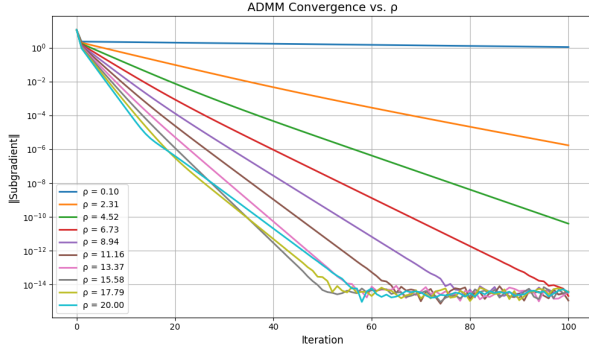


Fig. 8. State evolution of agents using CSS.

VI. DISCUSSION

The centralized approach achieves optimal consensus in one shot but requires global information. Among decentralized methods, ANM converges fastest due to its momentum term. ELC improves over CSS with adaptive steps, while CC and ADMM are more robust to problem variations but need parameter tuning.

VII. CONCLUSION

This report demonstrated decentralized consensus via optimization techniques. Agents cooperatively minimized a shared cost while adhering to local dynamics. Decentralized algorithms showed convergence to the centralized solution, with trade-offs in computation and communication.

REFERENCES

- [1] [Online]. Available: <https://jlmelville.github.io/mize/nesterov.html>
- [2] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, 2011.